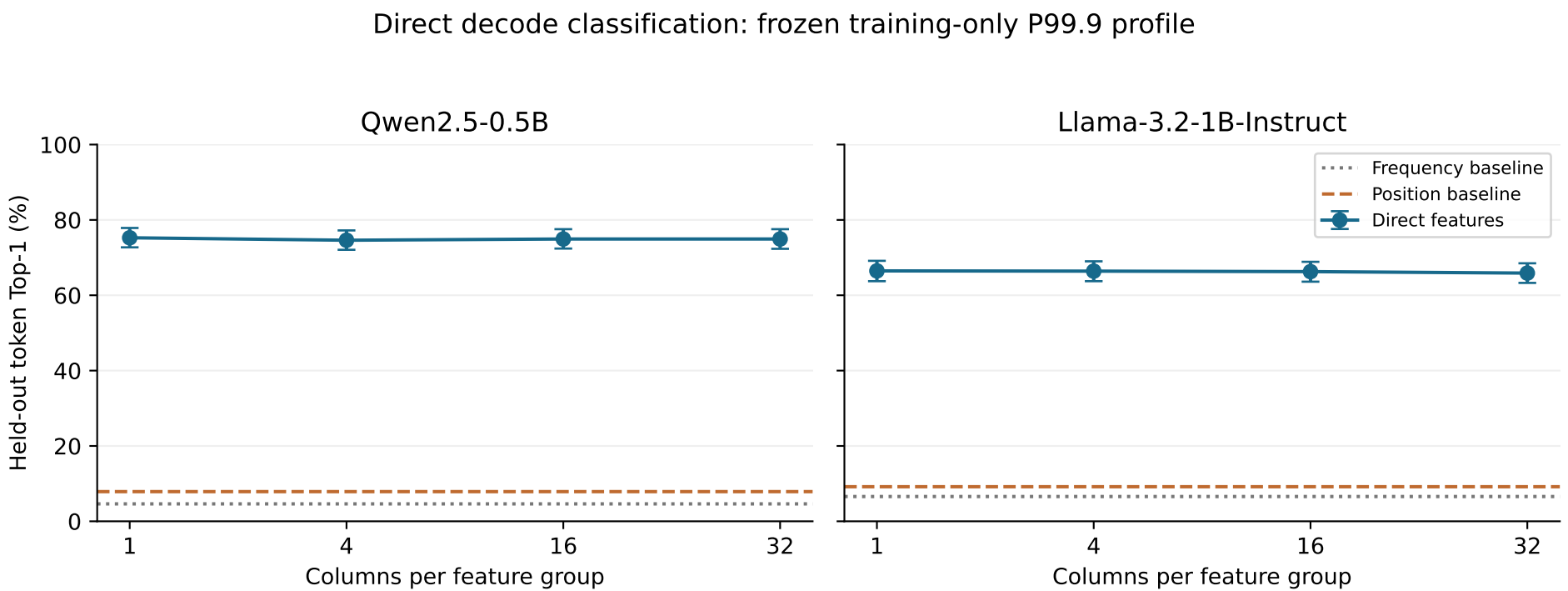
Direct decode classification: frozen training-only P99.9 profile
Qwen2.5-0.5B
Llama-3.2-1B-Instruct
Held-out token Top-1 (%)
100
80
60
40
20
0
1
4
16
32
Columns per feature group
Frequency baseline
Position baseline
Direct features
48 responses × up to 32 consumed tokens; response-bootstrap 95% intervals.